## A. Course Handout (Version 1.0)

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|---|---|---|---|
| Institute/School Name | Chitkara University Institute of Engineering and Technology | | |
| Department Name | Department of Computer Science & Engineering | | |
| Programme Name | Bachelor of Engineering (B.E.), Computer Science & Engineering | | |
| Course Name | System Design | Session | 2025-2026 |
| Course Code | 22CS024 | Semester/Batch | 5th/2023 |
| L-T-P (Per Week) | 2-0-2 | Course Credits | 03 |
| Pre-requisites | Software Engineering, Operating System, Computer Networks, DBMS | NHEQF Level | 5.5 |
| Course Coordinator | Dr. Suhasini | SDG Number | 4,8,9 |

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|---|---|
| **CLO01** | Gain a solid understanding of system design principles, advanced concepts, and architectural patterns, enabling the ability to design scalable, reliable, and maintainable systems that meet both functional and non-functional requirements |
| **CLO02** | Understand core database concepts, design principles, and normalization techniques, along with SQL operations and advanced topics enabling the ability to design efficient, consistent, and scalable database systems |
| **CLO03** | Learning and applying network communication models, protocols, and client-server architecture, along with key design principles and tools, enabling the development of secure and high-performance networked systems. |
| **CLO04** | Build a strong foundation in low-level system design by understanding operating system fundamentals, memory and cache management, and security mechanisms, enabling the development of optimized and secure system components |

### 1. Objectives of the Course

The course provides a wide scope of learning & understanding of the subject and the main objectives of the course are:

- To gain a solid understanding of system design principles, advanced concepts, and architectural patterns, enabling the ability to design scalable, reliable, and maintainable systems that meet both functional and non-functional requirements.
- To understand core database concepts, design principles, and normalization techniques, along with SQL operations and advanced topics enabling the ability to design efficient, consistent, and scalable database systems.
- To Learn and apply network communication models, protocols, and client-server architecture, along with key design principles and tools, enabling the development of secure and high-performance networked systems.
- To Build a strong foundation in low-level system design by understanding operating system fundamentals, memory and cache management, and security mechanisms, enabling the development of optimized and secure system components

### 2. Course Learning Outcomes: After completion of the course, student should be able to:

| | Course Learning Outcome | *POs | **CL | ***KC | Sessions |
|---|---|---|---|---|---|
| **CLO01** | Gain a solid understanding of system design principles, advanced concepts, and architectural patterns, enabling the ability to design scalable, reliable, and maintainable systems that meet both functional and non-functional requirements | PO1,PO2, PO3, PO4, PO9, PO11, PO12 | K6 | Factual, Conceptual, Procedural | 16 |

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| **CLO02** | Understand core database concepts, design principles, and normalization techniques, along with SQL operations and advanced topics enabling the ability to design efficient, consistent, and scalable database systems | PO1,PO2, PO3, PO4, PO9,PO10 PO11, PO12 | K5 | Factual, Conceptual, Procedural | 16 |
| **CLO03** | Learning and applying network communication models, protocols, and client-server architecture, along with key design principles and tools, enabling the development of secure and high-performance networked systems. | PO1,PO2, PO3, PO4, PO9,PO10 PO11, PO12 | K4 | Factual, Conceptual, Procedural | 16 |
| **CLO04** | Build a strong foundation in low-level system design by understanding operating system fundamentals, memory and cache management, and security mechanisms, enabling the development of optimized and secure system components | PO1,PO2, PO3, PO4, PO9,PO10 PO11, PO12 | K4 | Factual, Conceptual, Procedural | 16 |
| | **Total Contact Hours** | | | | 64 |

Revised Bloom's Taxonomy Terminology
* PO's available at (shorturl.at/cryzF)
**Cognitive Level =CL
***Knowledge Categories = KC

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 | PO8 | PO9 | PO10 | PO11 | PO12 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **CLO01** | H | H | H | H | | | | | M | M | L | L |
| **CLO02** | H | M | H | H | | | | | M | M | L | L |
| **CLO03** | H | H | M | M | | | | | L | L | L | L |
| **CLO04** | H | M | M | H | | | | | M | L | L | L |

H=High, M=Medium, L=Low

## 3. ERISE Grid Mapping

| Feature Enablement | Level (1-5, 5 being highest) |
|---|---|
| Entrepreneurship | 2 |
| Research | 4 |
| Innovation | 3 |
| Skills | 5 |
| Employability | 5 |

## 4. Recommended Books:

**B01.** System Analysis and Design, 7th edition, Julie E Kendall and Kenneth E Kendal, 2009
**B02**. Systems Analysis and Design, 9th edition, Gary Shelly, Harry J. Rosenblatt, 2011.
**B03.** System Analysis And Design, 5th edition, Wixom & Roth, 2012.
**B04.** Database System Concepts", 6th Edition by Abraham Silberschatz, Henry F. Korth, S. Sudarshan, McGraw-Hill, 2010.
**B05.** Data Communication and Networking, 4th Edition, Behrouz A. Forouzan, McGraw-Hill, 2007
**B06.** Operating System Concepts Essentials, 9th Edition by Avi Silberschatz, Peter Galvin, Greg Gagne, Wiley Asia Student Edition, 2013.

5. **Other readings and relevant websites:**

| S.N. | Link of Journals, Magazines, Websites, and Research Papers |
|---|---|
| 1. | nptel.ac.in/courses/106105175 |
| 2. | nptel.ac.in/courses/106106095 |
| 3. | nptel.ac.in/courses/106108101 |
| 4. | https://www.ece.uvic.ca/~itraore/elec567-13/notes/dist-03-4.pdf |
| 5. | https://www.udemy.com/course/system_design_lld_hld/ |
| 6. | https://arpitbhayani.me/masterclass/ |

6. **Recommended Tools and Platforms**

Canva, Draw.io, Diagrams.net, MS SQL Server/MySQL

7. **Course Plan:**

| Lecture No. | Topics | Recommended Books / Resources |
|---|---|---|
| 1-4 | **Fundamentals of System Design:** Introduction to System Design, System Design Characteristics**:** Availability, Reliability, Scalability: Vertical Scaling (Scaling Up), Horizontal Scaling (Scaling Out), Maintainability, Consistency, Fault Tolerance, Latency, Throughput<br>Comparison: Performance vs Scalability, Latency vs Throughput, Availability vs Consistency, Maintainability - Modular Design, Clean Code Principles, CI/CD Pipelines | B01,B02,B03 |
| 5-8 | **Advanced System Design Concepts**: CAP Theorem, Consistency Patterns: Weak Consistency, Eventual Consistency, Strong Consistency,<br>Functional and Non-Functional Requirements, Design Approaches: High Level Design, Low Level Design<br>Architectural Patterns-Monolithic Architecture, Microservices, Event-Driven Architecture | B01,B02,B03 |
| 9-12 | **Fundamentals of Database:** What is Database, Characteristics,<br>**Data Models**, Database Schema: Logical, Conceptual, Physical<br>**Data Integrity and Consistency:** Primary Key, Candidate Keys, Super Key, Check Constraint, Unique Constraint, Not Null Foreign Keys<br>**Database Design:** What is Database Design, Steps to Ensure Good Database Design, Database Design Lifecycle Phases: Requirement Analysis, Logical Design, Physical Design, Object Oriented Design Principles | B04 |
| 13-16 | Functional Dependencies, Normalization:1NF, 2NF, 3NF, BCNF, Transactions and ACID Properties, Indexing Recovery and Back up<br>**Recap of SQL:** Introduction, SQL Queries: DDL, DML and DCL, TCL and DQL, Nested and Correlated Queries<br>**Advanced Concepts :** Sharding, Replication,SQL vs NoSQL- Tradeoff. | B04 |

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|---|---|---|
| **ST1 (Syllabus Covered from Lecture 1 - 16)** | | |
| 17-20 | **Network Design:** Introduction to OSI, TCP/IP Model, Communication - Synchronous vs. Asynchronous Communication, Communication Protocols: HTTP, SMTP, SNMP, POP3, UDP, TCP, IP, RPC, REST, Graph QL and Web Sockets<br><br>**Client Server Design Architecture:** Introduction, Components : Client, Server, Network & Protocols, Middleware UI, Application Logic, Design Principles: Modularity, Scalability, Reliability and Availability, Load Balancing, Security, Performance Optimization, Interoperability and Maintainability | B05 |
| 21-24 | **Overview of Frameworks and Tools for Client-Server Development**: Server-Side Frameworks, Client-Side Frameworks<br><br>Domain Name System(DNS), Firewalls, Load Balancers: Application and Network, CDN (Content Delivery Networks: Push CDNs and Pull CDNs | B05 |
| 25-28 | **Low Level Design:** Introduction to OS and Types of OS, **Process Management:** PCB, Process Scheduling: Pre-emptive Scheduling and Non-pre-emptive**,** Race Conditions. **Memory Management:** Contiguous Memory Allocation, Paging, Segmentation. **Virtual Memory:** Thrashing and Page Faults, Page Replacement Algorithms. | B06 |
| **ST2 ( Syllabus Covered from Lecture 1-28)** | | |
| 29-32 | **Advanced Memory Management:** Caching, Cache Memory, Cache Replacement, Cache Eviction Policies (LRU, LFU), Distributed Caching (Redis, Memcached)<br>**Security:** Access Control and Authentication, Malware and Defence Mechanisms. Authentication and Authorization - OAuth 2.0, JWT (JSON Web Tokens), SSO (Single Sign-On) | B06 |
| **Additional Topics(Optional):** DevOps Concepts: CI/CD Pipelines: Continuous Integration, Monitoring and Logging: Prometheus, Grafana | | |
| **Practice Problems and Case Studies: Any one Case Study to be allotted for each group of students (3-4 students).**<br>• Design a URL Shortener (e.g., TinyURL)<br>• Design an E-commerce Platform(eg. Amazon)<br>• Design a Messaging Service (e.g., WhatsApp)<br>• Design a Social Media News Feed (e.g., Facebook or Twitter)<br>• Design a Ride-Sharing System (e.g., Uber or Lyft)<br>• Design a Video Streaming Service (e.g., YouTube or Netflix)<br>• Design a File Storage Service (e.g., Dropbox or Google Drive)<br>• Design a Notification System (e.g., Push Notifications or Email Alerts) | | |
| **Case Study Evaluation – Presentation of the Design of the Selected Problem** | | |
| **ETE (Full Syllabus)** | | |

## 8. Delivery/Instructional Resources

| Lecture Numbers | Topics | Web References | Audio-Video References |
|---|---|---|---|
| 1-4 | **Fundamentals of System Design:** Introduction to System Design, System Design Characteristics**:** Availability, Reliability, Scalability: Vertical Scaling (Scaling Up), Horizontal Scaling (Scaling Out), Maintainability, Consistency, Fault Tolerance, Latency, Throughput Comparison: Performance vs Scalability, Latency vs Throughput, Availability vs Consistency, Maintainability - Modular Design, Clean Code Principles, CI/CD Pipelines | https://www.geeksforgeeks.org/system-design/how-to-draw-high-level-design-diagram,<br><br>https://www.geeksforgeeks.org/system-design/system-design-tutorial/<br><br>https://www.designgurus.io/blog/system-design-interview-fundamentals | https://www.youtube.com/c/SystemDesignInterview,<br><br>https://www.youtube.com/watch?v=m8Icp_Cid5o,<br><br>https://www.youtube.com/watch?v=F2FmTdLtb_4 |
| 5-8 | **Advanced System Design Concepts**: CAP Theorem, Consistency Patterns: Weak Consistency, Eventual Consistency, Strong Consistency,<br>Functional and Non-Functional Requirements, Design Approaches: High Level Design, Low Level Design<br>Architectural Patterns-Monolithic Architecture, Microservices, Event-Driven Architecture | https://systemdesign.one/<br><br>https://www.geeksforgeeks.org/dbms/the-cap-theorem-in-dbms/ | https://www.youtube.com/watch?v=6YblqXh8nDY&t=2s |
| 9-12 | **Fundamentals of Database:** What is Database, Characteristics, **Data Models**, Database Schema: Logical, Conceptual, Physical<br>**Data Integrity and Consistency:** Primary Key, Candidate Keys, Super Key, Check Constraint, Unique Constraint, Not Null Foreign Keys<br>**Database Design:** What is Database Design, Steps to Ensure Good Database Design, Database Design Lifecycle Phases: Requirement Analysis, Logical Design, Physical Design, Object Oriented Design Principles | https://www.geeksforgeeks.org/dbms/getting-started-with-database-management-system/<br>https://www.geeksforgeeks.org/dbms/database-design-in-dbms/ | https://www.youtube.com/watch?v=kDiZkmAzevY&t=4s<br><br>https://www.youtube.com/watch?v=GH1KfNdn7ls&t=1s |
| 13-16 | Functional Dependencies, Normalization:1NF, 2NF, 3NF, BCNF, Transactions and ACID Properties, Indexing Recovery and Back up<br>**Recap of SQL:** Introduction, SQL Queries: DDL, DML and DCL, TCL and DQL, Nested and Correlated Queries<br>**Advanced Concepts :** Sharding, Replication,SQL vs NoSQL-Tradeoff. | https://www.geeksforgeeks.org/dbms/getting-started-with-database-management-system/<br>https://www.geeksforgeeks.org/dbms/database-recovery-techniques-in-dbms/ | https://www.youtube.com/watch?v=1aRhS-FM1IA<br><br>https://www.youtube.com/watch?v=EKG_OUraGDw |

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|---|---|---|---|
| 17-20 | **Network Design:** Introduction to OSI, TCP/IP Model, Communication - Synchronous vs. Asynchronous Communication, Communication Protocols: HTTP, SMTP, SNMP, POP3, UDP, TCP, IP, RPC, REST, Graph QL and Web Sockets<br><br>**Client Server Design Architecture:** Introduction, Components : Client, Server, Network & Protocols, Middleware UI, Application Logic, Design Principles: Modularity, Scalability, Reliability and Availability, Load Balancing, Security, Performance Optimization, Interoperability and Maintainability | https://medium.com/%40goldengrisha/api-approaches-you-should-know-rest-graphql-websockets-and-beyond-6d8e939f5c89<br><br>https://medium.com/tradeling/network-communication-concepts-f808c84afad7 | https://www.youtube.com/watch?v=8EpZOSf8CJE<br><br>https://www.youtube.com/watch?v=6fYsCkvuPbw<br><br>https://www.geeksforgeeks.org/system-design/client-server-architecture-system-design/ |
| 21-24 | **Overview of Frameworks and Tools for Client-Server Development**: Server-Side Frameworks, Client-Side Frameworks<br><br>Domain Name System(DNS), Firewalls, Load Balancers: Application and Network, CDN (Content Delivery Networks: Push CDNs and Pull CDNs | https://www.browserstack.com/guide/web-development-frameworks<br><br>https://www.splunk.com/en_us/blog/learn/cdns-vs-load-balancers.html | https://www.youtube.com/watch?v=bJ9NnLLMQ78<br><br>https://www.geeksforgeeks.org/videos/client-server-architecture-system-design/ |
| 25-28 | **Low Level Design:** Introduction to OS and Types of OS, **Process Management:** PCB, Process Scheduling: Pre-emptive Scheduling and Non-pre-emptive**,** Race Conditions. **Memory Management:** Contiguous Memory Allocation, Paging, Segmentation. **Virtual Memory:** Thrashing and Page Faults, Page Replacement Algorithms. | https://www.geeksforgeeks.org/operating-systems/operating-systems/ | https://www.youtube.com/watch?v=gTtfq610s48<br><br>https://www.youtube.com/watch?v=OtWvj4vTih0&t=1s |
| 29-32 | **Advanced Memory Management:** Caching, Cache Memory, Cache Replacement, Cache Eviction Policies (LRU, LFU), Distributed Caching (Redis, Memcached)<br>**Security:** Access Control and Authentication, Malware and Defence Mechanisms. Authentication and Authorization - OAuth 2.0, JWT (JSON Web Tokens), SSO (Single Sign-On) | https://www.geeksforgeeks.org/operating-systems/operating-systems/<br><br>https://www.geeksforgeeks.org/system-design/cache-eviction-policies-system-design/<br><br>https://algocademy.com/blog/cache-algorithms-fifo-vs-lru-vs-lfu-a- | https://www.youtube.com/watch?v=xY7cQNF8awA<br><br>https://www.youtube.com/watch?v=ZqI7i3v4baQ<br><br>https://www.youtube.com/watch?v=ZDuRmhLSLOY<br><br>https://www.youtube.com/watch?v=xJA8tP74KD0 |

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| | | comprehensive-guide/<br><br>https://www.geeksforgeeks.org/spring-security-oauth2-vs-jwtjson-web-tokens/<br><br>https://www.tutorialspoint.com/spring_security/spring_security_with_jwt.htm | |

9. **Lab Plan**

| Lab No. | Experiments | Learning Resources |
|---|---|---|
| Week 2 | Design Approaches: High Level Design, Low Level Design using Designing Tools like Canva, Drawio or any other tool | Draw.io / diagrams.net, https://www.canva.com/en_in/, www.geeksforgeeks.org |
| Week 4 | SQL Queries: DDL, DML and DCL,TCL and DQL, Nested and Correlated Queries | https://www.mygreatlearning.com/blog/sql-commands/ |
| Week 5-16 | **Practice Problems and Case Studies: Any one Case Study to be allotted for each group of students (3-4 students).**<br>• Design a URL Shortener (e.g., TinyURL)<br>• Design an E-commerce Platform(eg. Amazon)<br>• Design a Messaging Service (e.g., WhatsApp)<br>• Design a Social Media News Feed (e.g., Facebook or Twitter)<br>• Design a Ride-Sharing System (e.g., Uber or Lyft)<br>• Design a Video Streaming Service (e.g., YouTube or Netflix)<br>• Design a File Storage Service (e.g., Dropbox or Google Drive)<br>• Design a Notification System (e.g., Push Notifications or Email Alerts) | |

****Note: Practice Problems and Case Studies mentioned above must be assigned to the group of 3-4 students in Week 5 and Evaluation will be done at the end of the semester before ETE.**

10. **Action plan for different types of learners**

| Slow Learners | Average Learners | Fast Learners |
|---|---|---|

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|---|---|---|
| • Remedial Classes on Saturdays<br>• Encouragement for improvement using Peer Tutoring<br>• Use of Audio and Visual Materials<br>• Use of Real-Life Examples | • Workshops<br>• Formative Exercises used to highlight concepts and notions<br>• E-notes and E-exercises to read ahead of the pedagogic material. | • Engaging students to hold hands of slow learners by creating a Peer Tutoring Group<br>• Design solutions for complex problems<br>• Design solutions for complex problems<br>• Presentation on topics beyond those covered in CHO |

11. **Evaluation Scheme & Components:** As per Choice Based Credit System(CBCS) , student can choose any one evaluation scheme given below:

| Evaluation Component | Type of Component | No. of Assessments | Weightage of Component | Mode of Assessment |
|---|---|---|---|---|
| **Component 1** | Formative Assessment (FA) | 01# | 10% | Online(On-Campus) |
| **Component 2** | Sessional Tests (STs) | 02* | 30% | Online(On-Campus) |
| **Component 3** | End Term Examination | 01 | 60% | Online(On-Campus) |
| **Total** | | **100%** | | |

* Out of 02 STs, Best 1 ST for final marks evaluation of STs will be considered.
#FA evaluation will be **Case Study Evaluation – Presentation of the Design of the Selected Problem**
**As per Academic Guidelines, a minimum of 75% attendance is required to become eligible for appearing in the End Semester Examination.

12. **Syllabus of the Course:**

| **Subject:** System Design | | |
|---|---|---|
| **Topic (s)** | **No. of Sessions** | **Weightage %** |
| **Fundamentals of System Design:** Introduction to System Design, System Design Characteristics**:** Availability, Reliability, Scalability: Vertical Scaling (Scaling Up), Horizontal Scaling (Scaling Out), Maintainability, Consistency, Fault Tolerance, Latency, Throughput<br>Comparison: Performance vs Scalability, Latency vs Throughput, Availability vs Consistency, Maintainability - Modular Design, Clean Code Principles, CI/CD Pipelines<br><br>**Advanced System Design Concepts**: CAP Theorem, Consistency Patterns: Weak Consistency, Eventual Consistency, Strong Consistency, | **16** | **25%** |

| | | |
|---|---|---|
| Functional and Non-Functional Requirements, Design Design Approaches: High Level Design, Low Level Design, Architectural Patterns-Monolithic Architecture, Microservices, Event-Driven Architecture | | |
| **Fundamentals of Database:** What is Database, Characteristics,<br>**Data Models**, Database Schema: Logical, Conceptual, Physical<br>**Data Integrity and Consistency:** Primary Key, Candidate Keys, Super Key, Check Constraint, Unique Constraint, Not Null Foreign Keys<br>**Database Design:** What is Database Design, Steps to Ensure Good Database Design, Database Design Lifecycle Phases: Requirement Analysis, Logical Design, Physical Design, Object Oriented Design Principles<br><br>Functional Dependencies, Normalization:1NF, 2NF, 3NF, BCNF, Transactions and ACID Properties, Indexing Recovery and Back up<br>**Recap of SQL:** Introduction, SQL Queries: DDL, DML and DCL, TCL and DQL, Nested and Correlated Queries<br><br>**Advanced Concepts :** Sharding, Replication,SQL vs NoSQL- Tradeoff. | **16** | **25%** |
| **Network Design:** Introduction to OSI, TCP/IP Model, Communication - Synchronous vs. Asynchronous Communication, Communication<br>Protocols: HTTP, SMTP, SNMP, POP3, UDP, TCP, IP, RPC, REST, Graph QL and Web Sockets | **16** | **25%** |

| | | |
|---|---|---|
| **Client Server Design Architecture:** Introduction, Components : Client, Server, Network & Protocols, Middleware UI, Application Logic, Design Principles: Modularity, Scalability, Reliability and Availability, Load Balancing, Security, Performance Optimization, Interoperability and Maintainability<br><br>**Overview of Frameworks and Tools for Client-Server Development**: Server-Side Frameworks, Client-Side Frameworks, Domain Name System(DNS), Firewalls, Load Balancers: Application and Network, CDN (Content Delivery Networks: Push CDNs and Pull CDNs | | |
| **Low Level Design:** Introduction to OS and Types of OS, **Process Management:** PCB, Process Scheduling: Pre-emptive Scheduling and Non-pre-emptive**,** Race Conditions. **Memory Management:** Contiguous Memory Allocation, Paging, Segmentation. **Virtual Memory:** Thrashing and Page Faults, Page Replacement Algorithms. **Advanced Memory Management:** Caching, Cache Memory, Cache Replacement, Cache Eviction Policies (LRU, LFU), Distributed Caching (Redis, Memcached) **Security:** Access Control and Authentication, Malware and Defence Mechanisms. Authentication and Authorization - OAuth 2.0, JWT (JSON Web Tokens), SSO (Single Sign-On) | **16** | **25%** |

**This Document is approved by:**

| **Designation** | **Name** | **Signature** |
|---|---|---|
| **Course Coordinator** | Dr. Suhasini | |
| **Head-Academic Delivery** | Dr. Susheela Hooda | |
| **Dean** | Dr. Rupali Gill | |
| **Date** | 24.06.2025 | |